by the within-tolerance variations of the actual resistors used. Switch S1 is a three-way rotary switch which is used to connect two sections in parallel for the required dummy load. As the switch is rated at 6 A per section, it should handle up to 12 A quite reliably.

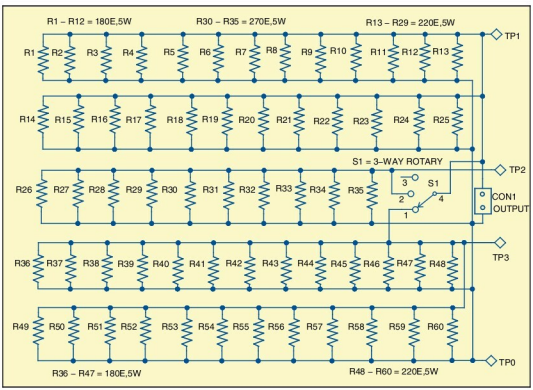


Fig. 3: Complete circuit diagram of dummy load

Construction and testing

Since the dummy load can be constructed more easily using general-purpose PCBs, as shown in Fig. 1, so no PCB design is provided.

Assemble three sets of resistor banks using back-to-back general-purpose PCBs as shown in Fig. 1. The first set is built using resistors R1 through R25 for making a load of about 8 Ω. The second set is built with resistors R26 through R35 for making a dummy load of about 24 Ω. The third set is built using resistors R36 through R60 for making a dummy load of about 8 Ω.

Connect switch S1 and the loads as shown in Fig. 2. After proper assembly, insert the fan in such a way that it can easily dissipate the heat. For testing, use a multimeter to measure the resistance according to the test-point table. ●

The author is a regular contributor to EFY and has many articles published to his credit in India and abroad



